



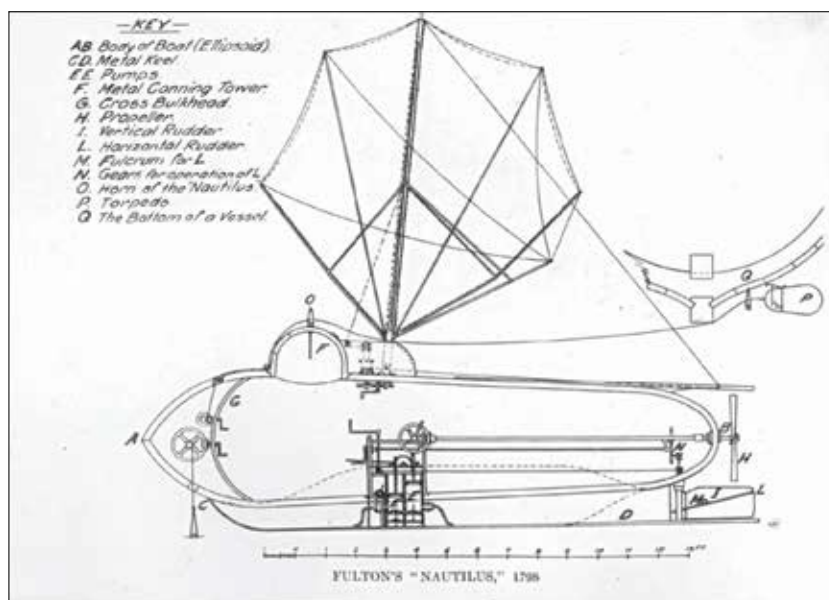
New tech to help stroke patients

A new nerve stimulation technology can help 'rewire' the brains of stroke patients to regain use of their arms. <https://digit.in/jun18-A27-1>



Tesla reduced grid service costs

Tesla's giant battery in Australia has effectively reduced the grid service charges by a whopping 90%. <https://digit.in/june18-A27-2>



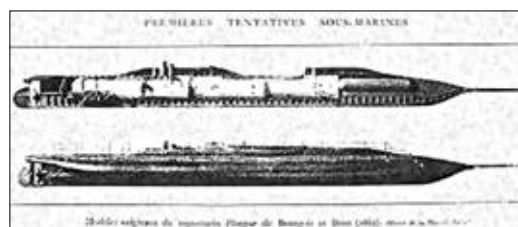
Fulton's Nautilus

it robust, but the design was further improved after intervention from Swedish Industrialist Thorsten Nordenfelt. Their combined efforts led to a series of Nordenfelt submarines that achieved previously unaccomplished feats. Their success drew even more military attention to submarines.

The Holland Type VI submarine, designed in 1896, used a combustion engine on the surface and battery power when submerged. While it was not the first electricity powered submarine, it did end up being the first submarine commissioned by the United States Navy in 1900, renamed to USS Holland. The Holland class submarine would gain in popularity and be adapted all over the world, until the start of World War I would push submarine research to a new level.

THE MODERN SUBMARINE AND WARTIME DEVELOPMENTS

In 1914, the Royal Navy (Britain) had



Plongeur

the largest submarine fleet in service. They were designed with a range of 2500 nautical miles, better living conditions for the crew and gun decks. They were fitted with twin screws for greater manoeuvrability and with innovative saddle tanks.

They also had wireless transmitters. Its profile resembled the modern submarine very closely.

During the interwar years, various new submarine profiles were developed. Perhaps the most interesting among these is the submarine aircraft carrier. The British HMS M2, followed by the French Surcouf, and numerous aircraft-carrying submarines in the Imperial Japanese Navy are notable examples as well.

Early design during these times combined the efforts of a diesel engine and an electric motor on the same shaft to use the former for charging the batteries and the latter to drive the submarine underwater. Eventually, they were separated, allowing greater flexibility to submarines in terms of maneuverability.

NUCLEAR SUBMARINES

The first nuclear powered submarine was the USS Nautilus, commissioned in 1954. The advantage of using nuclear power is that the submarine could stay underwater for nearly unlimited time. Most submarines built ever since have been nuclear powered. Today, six countries deploy some form of nuclear-powered strategic submarines: the United States, Russia, France, the United Kingdom, China, and India.

THE FUTURE

The recent German 212 type submarine was the first to use hydrogen fuel cells for propulsion, powered by nine 34-kilowatt fuel cells. Multiple smaller commercial submarines that do not have a need to be operated independently, often run on small batteries that are recharged by the ship. Introduction of the quieter pump-jet propulsors also gave a tactical advantage in the 20th century.



HMS M2 launches a specially designed Parnall Peto seaplane. It sank accidentally in 1932

The future of submarine propulsion lies with the magnetohydrodynamic drive. A magnetohydrodynamic drive or MHD accelerator is a method for propelling vehicles using only electric and magnetic fields with no moving parts, accelerating an electrically conductive propellant (liquid or gas). As the liquid moves to the back of the vehicle, it is propelled forward as a reaction. While this design is currently impractical due to the low speeds it generates, it has been quite popular in fictional submarines - a notable example being the one in the film adaptation of *The Hunt for Red October*. 